Trend Prediction of GitHub using Time Series Analysis

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***Abstract*—GitHub is a web-based collaborative platform for software developers with around 28 million users and 57 million public repositories which promote social coding. GitHub is an evolving network with the developments in the projects and repositories. The underlying working principle of GitHub is based on the events occurring in a time-based manner. The number of events of a repository is proportional to its popularity. In this proposed work, we perform trend prediction on GitHub. The trend prediction is categorised into three tasks, repository trend prediction, language trend prediction and domain trend prediction respectively. Using most relevant and recurring events, such as Create, Fork, Pullrequest, Push and issue, a multivariate time series is constructed. Trends are predicted with time series forecasting using Long Short Term Memory(LSTM) model. The trend prediction results compare the repositories, languages and domains and help in identifying the trending ones. GitHub insights are taken to facilitate social coding and in selecting the relevant languages and domains for the upcoming projects.**

***Index Terms*—GitHub, time series analysis, Long short term memory(LSTM), trend prediction, forecasting**

I. Introduction

Social networking platforms are the main resource for the creation and sharing of information and ideas. GitHub is one of those popular social networking platforms with a large number of users. GitHub extends numerous features and functionalities which excites and entices users across the world. It enables the developers to collaborate on projects more efficiently and effectively by providing several features. Being a version controlled system based on many events, the data is cached in a time-based manner. There are more than 45 events in GitHub which assists users in the development of their projects. The variation in the trends in technology is deeply reflected in this platform. Numerous amount of works have been focused on mining GitHub [1] [2], an online code hosting platform that relies on Git and additionally provides collaborative and social features.

Being an event-based data in a time-based manner, GitHub can be explored for various patterns that can provide insights which hold great applicability in various technical fields. This can be utilised for analysis of trending techniques to apply in the forthcoming projects. Repositories are defined as the containers where the project and its files are stored. The evolving repositories show their developments over time through their growing popularity so as the languages. The domain adopted for solving the problem in the repositories also gains its popularity by the increasing application to address various problems along with the repositories over time. The proposed work is about analysing the GitHub data to model the evolving repositories and the dynamic trends in the programming language and domains.

In this work, we apply time series analysis over GitHub due to its evolving nature. Eventhough there are various traditional models, the non-linearity of the time series invites the need for deep architectures for its analysis. Recurrent Neural Network(RNN) is a deep neural network specifically used for the non-linear time series analysis. It is designed for recognizing patterns in sequence data such as numerical time series data. Eventhough these RNNs worked to some extent, they had a rather larger downfall as they lead to vanishing gradient problem which demands the requirement of some new model for solving real-time problems. This is where Long Short Term Memory (LSTM) neural network came to the picture. Like RNN neurons, LSTM neurons kept a context of memory for tackling sequential and temporal problems without the issue of vanishing gradient affecting their performance.

In the next two sections, we will present background and related work. In section IV , we present our proposed work for the trend prediction. Section V presents our experiments and the findings. Finally, we conclude this paper in section VI.

II. Background

A. GitHub

GitHub is the most popular web-based collaborative platform for software developers. Being started from 2008, it is the world's largest open resource software platform with over 28 million users and 57 million public repositories which facilitates social coding by providing a web interface to the git repository and tools for managing the collaboration. The source code and the complete history of all revisions done in that code are stored here. The members can follow, rate and fork each other's work. While studying through the evolution, it has been observed that GitHub become a highly evolving network by the dynamic developments in the repositories. The changing trends in technology and language are reflected in the GitHub platform. Mining the data from GitHub can

bring many insights which help in various fields including the details about the changing trends in technology and domains. Instead of being stored as repository wise, it has emerged into an event-based one with an increasing number of users and repositories. With more than 45 events, now the working of GitHub is fully based on these events. A few of them being frequent helps the repositories to become popular and more trending. In this paper, the data from 2015 to 2018 is analysed. The trend prediction is performed using some of the important and frequent events such as fork event, Pullrequestevent, Push event etc...

The evolution of GitHub is based on the advancements in the repositories which in turn reflects the changing trends in the technologies. The main contribution of this paper is the modelling and trend prediction of repositories, languages and domains based on the frequent events.

B. Time Series Analysis

Time Series is a sequence of well-defined data points taken at equally spaced intervals of time. Large data size, high dimensionality and continuous updating are the key features of time series data. Time series analysis is the process of using statistical methods to analyse time series data. Through the analysis, meaningful statistics and characteristics of the data are extracted. This helps in understanding underlying forces that lead to various trends in time series data points.

Time series can be categorized into univariate time series and multivariate time series based on the number of features required for the analysis and prediction. of time series. Further time series can also be divided into linear time series and non-linear time series based on the behaviour of time series. A univariate, real-valued time series, defined as x = x (n): n =1,..., N, is an ordered set of N real-valued variables. Here single feature values are observed sequentially through time. And results depends on observation value of single feature. A multivariate time series is considered when multiple features influence prediction result for more accurate prediction.

Various models were introduced for time series analysis. Amongst these, Recurrent Neural Networks gained wide acceptance. Later, due to the vanishing gradient problem, a more efficient model Long Short Term Memory(LSTM) was introduced to counterpart the RNN problem. The LSTM is a standard and well-known Recurrent Neural Network. They can model long-term dependencies better than Recurrent Neural Network, which makes them more preferable for time series analysis and prediction problems. LSTM has a sophisticated structure which is made up of gates including a forget gate network, output gate network. These gates control the memory cell and regulate the flow of information to hidden neurons [3]. Forget gate network manages the flow of information of internal cell to the next time step. Input gate network scales the input block to the internal cell. And the output gate network controls the output of the internal cell at each time step. Hence the updation of the internal cell is said to be nearly linear, and the gradient is propagated easily, so the memory cell can retain state over long periods of time. Depending on the memory cell and the set of gate networks, LSTM can represent long-term or short-term dependencies in sequential data more effectively.

C. Problem Statement

GitHub is an evolving system with around 28 million users and 57 million public repositories. The evolution of GitHub is based on the advancements in the repositories which in turn reflects the changing trends in the booming technologies.

III. Related Work

GitHub facilitates social coding by providing a web interface to the git repository and management tools for collaboration. GitHub is one of the popular open source platforms which attracts the attention of research scholars. The popularity of the repositories was predicted using the multiple linear regression model using star event by [2], [4]. There are various methods for reviewer recommendation [5]. The association rules mined from the git repositories are used to find factors influencing the assignment of reviewers to pullrequests [6]. One of the main challenges in GitHub is to find the user's social influence on the platform. Later, a following-star-fork activity-based approach was adapted for measuring influence [1]. Popular users influence their followers by guiding them to new projects. It has been observed that the influence rate increases in proportional to the popularity of users [7]. Potential impact factors are defined and its correlation with the unfollowing behaviour of users [8]. Factors that affect the issue closure rates in open source software projects based on the short term and long term issue closure is analysed in [9]. [10] focuses on various types of programming languages and its uses in GitHub. Influence of GitHub repositories can be predicted based on the relation between the users and repositories [11].

LSTM model has acquired wide acceptance in several fields due to its efficiency in solving sequential and temporal problems without the issue of vanishing gradient affecting their performance. [12] has applied the LSTM model for the real-time prediction of online purchases. Similarly, daily box office and stock exchange prediction are implemented efficiently using LSTM [13], [14]. LSTM model based prediction are also adapted in vegetation dynamics [15]. LSTM with Adam optimizer with adaptive learning rates is used for the tourism flow prediction [16]. A hybrid model of Convolutional Neural Network(CNN) along with the LSTM is used in the prediction of online reliability of the Service-oriented Systems [17]. LSTM with the Recurrent Neural Network can also be used for ad click prediction which influences the business to a great extent [18]. LSTM along with CNN also finds its applications in health prediction problems such as the seizure prediction problem [19]. Slow convergence and high sensitivity to the initial settings of the network are the problems of LSTM which leads to the introduction of adaptive LSTM network, where metaheuristic algorithms are used to train LSTM network through determining optimum values for its bias and weights [20].

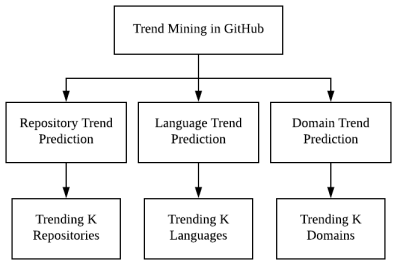

Fig. 1. Overall System model

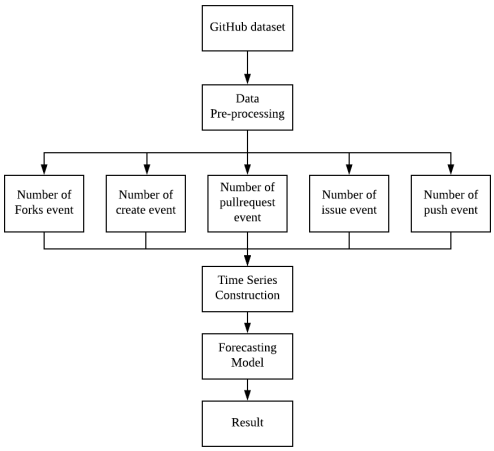

Fig. 2. System Model

IV. Proposed Work

This section describes the proposed method for modelling and predicting trends in GitHub. The problem of Mining data from GitHub and predicting trend is categorized into three subproblems as shown in fig. 1 which are Repository trend prediction, language trend prediction and domain trend prediction. The data extracted after pre-processing will have the number of frequent events along with the language and domain of 9350 popular repositories. As shown in the fig. 2 these pre-processed data is utilised for constructing the time series. LSTM model is applied for forecasting the trend using time series data. The efficient modelling of long term dependency makes the LSTM model best fit for prediction in the proposed system. Each trend prediction model has distinct pre-processed data for the time series construction.

A. Repository Trend Prediction

A repository is a digital storage space where projects and its files are stored. The increasing number of events in a repository accounts for the increasing popularity of the repository. From a large number of events, we consider four events for the repository trend prediction which are Fork event, pullrequest event, push event and issue event. Along with the number of pullrequest event, the number of pullrequestreview event and pullrequestreviewcomment event are also taken. Similarly, the number of issuecomment event is also considered along with the number of issue event. In repository trend prediction more weightage is given to fork and pullrequest event rather than issue event and push event. Using the count of these events a score is calculated in each time step using 1.

TABLE I
Multivariate time series for the repository id: 8514

Time step	Fork	Push	issue	pullrequest	Score
1	44	233	94	1	78.9
2	45	196	78	0	68.3
3	46	156	87	7	64.5
4	25	112	71	0	44.1
5	20	142	59	3	47.1

TABLE II
Multivariate time series for java

Time step	Create	Fork	Push	issue	pullrequest	Score
1	542	2821	3919	2482	535	1732.25
2	205	1829	1240	792	215	686.625
3	392	5793	6347	3586	994	2760.5
4	289	1771	2591	1169	306	986.25
5	20	469	759	702	41	344.25

$$score_t = \Sigma_i W_i X_i \tag{1}$$

where W_i is the weight of the event where values are taken emperically. X_i is the number of events occurring in the time period. Multivariate time series is constructed using the events along with the computed score. Table I has shown a sample time series for the repository id 8514. The LSTM model is used to forecast the score which is utilized for the trend prediction.

B. Language Trend prediction

There are various programming languages which gain and loses their popularity over time. We have considered about 11 languages for the study which are C, C++, Go, Java, JavaScript, PHP, Python, R, Ruby, Rust and Scala for this language prediction in GitHub. Create event, Fork event, Pullrequest event, push event and issue event are the factors used for the trend prediction. Create event, Fork event and pullrequest event have more relevance than the more frequent push event and issue event.Using the count of these events a score is calculated in each time step using 2.

$$score_t = \Sigma_i W_i X_i \tag{2}$$

where W_i is the weight of the event where values are taken emperically. X_i is the number of events occurred in the time period. Multivariate time series is constructed using the events along with the computed score. The time series is constructed for every language and time series for java is shown in Table II. Further, LSTM predicts the future score which is analyzed for the trend prediction.

TABLE III
MULTIVARIATE TIME SERIES FOR MACHINE LEARNING

Time step	Createt	Fork	Push	issue	pullrequest	Score
1	164	5769	1909	1131	131	1316.25
2	145	6785	1142	1016	152	1319.125
3	315	9466	3826	2095	405	2365.25
4	329	3220	1065	734	175	845.125
5	23	2185	782	534	107	536.875

C. Domain Trend Prediction

The projects are created in various domains based on the different problems arising over time. The creation and use of repositories of each domain defines the trend. Big data, cloud computing, deep-learning, image processing, IoT, machine learning Information Security and Web mining are the domains considered in this work. The labels extracted in pre-processing is used to classify the repositories into their domains. A score is generated using Create event, Fork event, pullrequest event, push event and issue events using 3.

$$score_t = \Sigma_i W_i X_i \tag{3}$$

where W_i is the weight of the event where values are taken emperically. X_i is the number of events occurred in the time period. A multivariate time series is constructed for each one of the domains using these events along with the generated score. A sample time series for machine learning is shown in Table III. LSTM predicts the score of each domain accurately which is analysed for calculating the trend.

V. EXPERIMENTS

A. Dataset

GitHub is a platform which works based on more than 45 events. The events includes CreateEvent, DeleteEvent, DeploymentEvent, DownloadEvent, FollowEvent, ForkEvent, GollumEvent, IssuesEvent, LabelEvent, MetaEvent, PullRequestEvent, ReleaseEvent, StarEvent etc.. We analyses the data from the year 2015 to 2018. Among the numerous events, only the frequent events are taken as the factors for the modelling and trend prediction. During pre-processing initial data are modified into the required format. Here, the count of event occurrence in each repository is extracted in pre-processing stage. The pre-processing stage consists of 5 steps:

1) Event based data of 2015, 2016, 2017 and 2018 from GH Archive (https://www.gharchive.org/) are collected
2) The number of times each event occurs in a repository is counted.
3) The active repositories are selected on the basis of the number of events.
4) The language and labels is extracted from the repositories.
5) The repositories are classified into different domains using SVM Classifier with the help of labels.

After pre-processing, the popular 93750 repositories are extracted and utilised for time series construction.

TABLE IV
REPOSITORY PREDICTION PERFORMANCE

Repositoryid	RMSE	R^2
34725861	0.001	0.999
247278	0.032	0.942
10969551	0.013	0.996
16286131	0.054	0.787
8514	0.074	0.214
258750	0.054	0.432
130013	0.167	0.461

TABLE V
LANGUAGE PREDICTION PERFORMANCE

Language	RMSE	R^2
C	0.011	0.883
C++	0.091	0.712
Go	0.031	0.651
Java	0.003	0.946
JavaScript	0.069	0.780
PHP	0.037	0.758
Python	0.013	0.661
R	0.085	0.153
Ruby	0.001	0.999
Rust	0.077	0.299
Scala	0.021	0.859

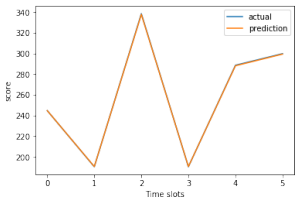

Fig. 3. Repository id: 34725861

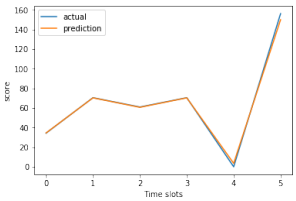

Fig. 4. Repository id: 10969551

B. Experimental Setup

Experimental setup for the proposed work is carried out on Ubuntu 16.04 64 bit operating system with a memory capacity of 15.6 GiB. Python 2.7, Anaconda, Tensorflow, Keras, Pandas and matplotlib are the tools used for implementation.

C. Experimental Results

1) Performance of the system model: The trend prediction problem can be divided in to three sub problems. The performance of the experiments conducted on these sub problems is measured in terms of Root Mean Square Error(RMSE) value and the coefficient of determination(R^2) value. RMSE being the error value and R^2 gives the correctness percent, makes the performance evaluation of the experiments efficient.

The Repository prediction performance is shown in Table IV and the performance of two sample repositories is shown in fig.4 and fig. 3. Table V shows the prediction performance of the language. Fig. 5 and fig. 6 shows the prediction performance of language java and ruby. The domain prediction performance is shown in Table VI along with the prediction performance of two sample domains big data and image processing shown in fig.7 and fig. 8. The models have shown good performance as the very low RMSE value and the high R^2 value which lies between 0 and 1.

2) Trending Repositories in GitHub: Various repositories in different domains gain popularity over time than the rest of the repositories. The idea of gathering and sharing of information is done through these repositories. According to the analysis in Table VII it has observed that various repositories shows hype in trends over time in various fields. The Table VIII shows the trending repositories in the year 2018. It has been observed that the repositories of the domains web mining, machine learning

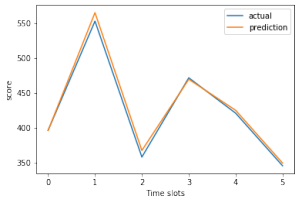

Fig. 5. Java

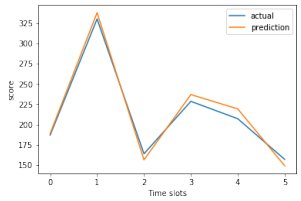

Fig. 6. Ruby

TABLE VI
DOMAIN PREDICTION PERFORMANCE

Domain	RMSE	R^2
Big Data	0.006	0.990
Cloud	0.006	0.969
Deep Learning	0.092	0.667
Image Processing	0.002	0.921
IoT	0.081	0.872
Machine Learning	0.066	0.418
Information Security	0.019	0.912
Web Mining	0.055	0.851

and information security gaining more popularity. Fig. 9 has shown the variation in the trend of some repositories over time.

3) Trending Languages in GitHub: The languages gain popularity according to the increase in the usage owing to their application in various fields. The features of each language greatly influence its trend. When mining GitHub we can see that the trend in the language also reflects in the GitHub platform also. Fig.10 has shown the analysis of the trending languages over the four years 2015,2016,2017 and 2018. According to the analysis, it has seen that javascript is the trending language over time. Java also trends which reflects the increasing use of java in various projects across various fields. When coming to the trend in 2018, as shown in fig. 11 JavaScript and java tops this time period also. Both Javascript and Java are the languages which consistently trending in GitHub. Fig.12 shows the variation of some languages over time periods 2015, 2016, 2017 and 2018.

4) Trending Domains in GitHub: The overall trend prediction result is shown in fig. 13. Machine Learning being the trending domain shows the increasing number of projects in the area of machine learning. Information security is also trending with the importance of security in all fields. When compared with the trend results of 2018 (shown in fig. 14), it has seen that cloud and big data are increasing its usage in 2018. Web mining is also an area with increased importance in the year 2018. The deep-learning domain gained its popularity over 2018 which shows its increasing application in projects. Fig. 15 has shown the variation of the trend in some domains over the four years.

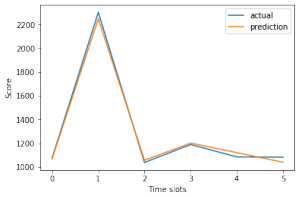

Fig. 7. Big Data

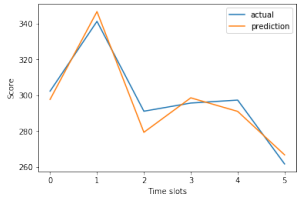

Fig. 8. Image Processing

TABLE VII
TRENDING REPOSITORIES OVER 2015, 2016, 2017, 2018

Repositoryid	Name	Domain
8514	rails/rails	Web mining
10969551	cms-sw/cmssw	web mining
34725861	everypolitician/everypolitician-data	Web mining
16286131	mapbox/mapbox-gl-native	Web mining
258750	theforeman/foreman	Cloud Computing
108278	chef/chef	Information Security
130013	akka/akka	Cloud Computing

TABLE VIII
TRENDING REPOSITORIES IN 2018

Repositoryid	Name	Domain
53493223	BrewTestBot/homebrew-core	Information Security
47203045	DataDog/integrations-core	Machine Learning
7548986	laravel/framework	Web mining
1390248	saltstack/salt	Cloud Computing
87395542	alpeware/chrome-headless-trunk	Web mining
49575911	sestegra/linuxbrew-dart	Information Security
142234428	triggermesh/tm	Cloud Computing

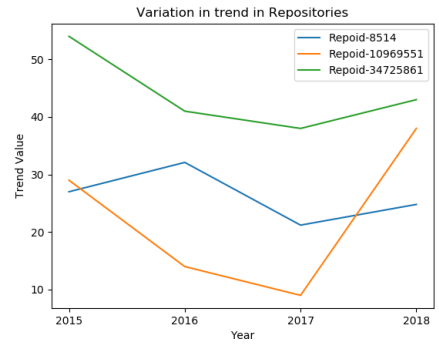

Fig. 9. variation in trend of some repositories

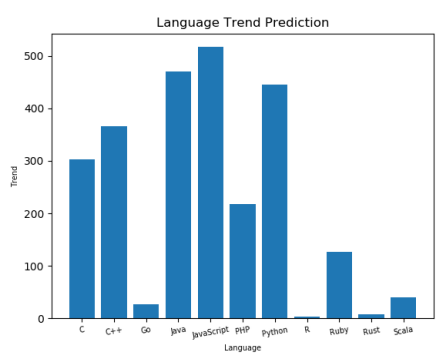

Fig. 10. Language Trend prediction over 2015, 2016, 2017, 2018

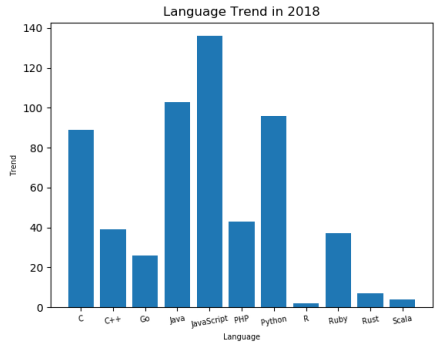

Fig. 11. Language trend in 2018

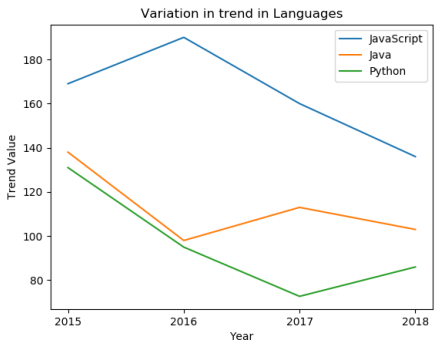

Fig. 12. Variation of trend in Language

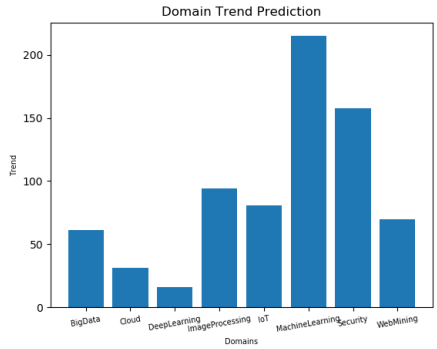

Fig. 13. Domain Trend Prediction over 2015, 2016, 2017, 2018

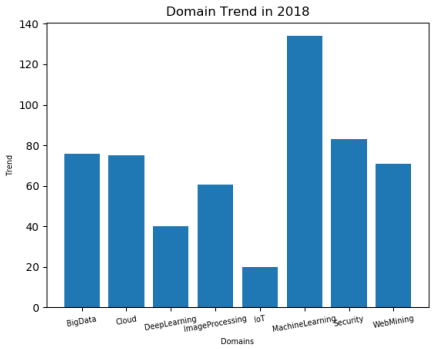

Fig. 14. Domain Trend in 2018

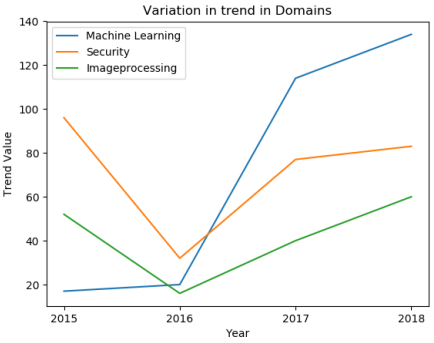

Fig. 15. Variation in trend in Domains

VI. Conclusion

In this paper, we propose a model for predicting the trend in GitHub. Unlike the existing systems which take only single events for predicting the popularity of repository, we use the count of five events which occur frequently and are more relevant for prediction. Create, fork, pullrequest, push and issue events are used for the trend prediction. The prediction problem is classified into three subproblems namely repository, language and domain trend prediction. A multivariate time series is constructed using these events which is used for forecasting the trend. LSTM model is adapted for the forecasting due to its efficiency in modelling long term dependency in non-linear time series. Experimental results proves that the proposed system gives a better overview of various events happening over GitHub. It also helps in selecting the trending ones for your upcoming projects.

The main limitations experienced during this work was huge datasize and its processing time. Big data analytics platform can be used for processing huge dataset to overcome this limitation and hence can serve as a future extension to this paper. In future work, we can also build a user trend prediction which will help in understanding the user's behaviour and prediction of upcoming projects. Adapting the trend prediction of GitHub into a big data analytics platform is also a challenge need to be solved. Exploring the network structure of GitHub and applying social network analysis methods for trend prediction can be another interesting direction of future work.

Acknowledgment

The infrastructure used for conducting this work is funded by FIST which is sanctioned by DST to NSS College of Engineering, Palakkad. We would like to express our gratitude to the Department of Computer Science and Engineering, N.S.S College of Engineering, Palakkad for providing the required computing infrastructure.

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